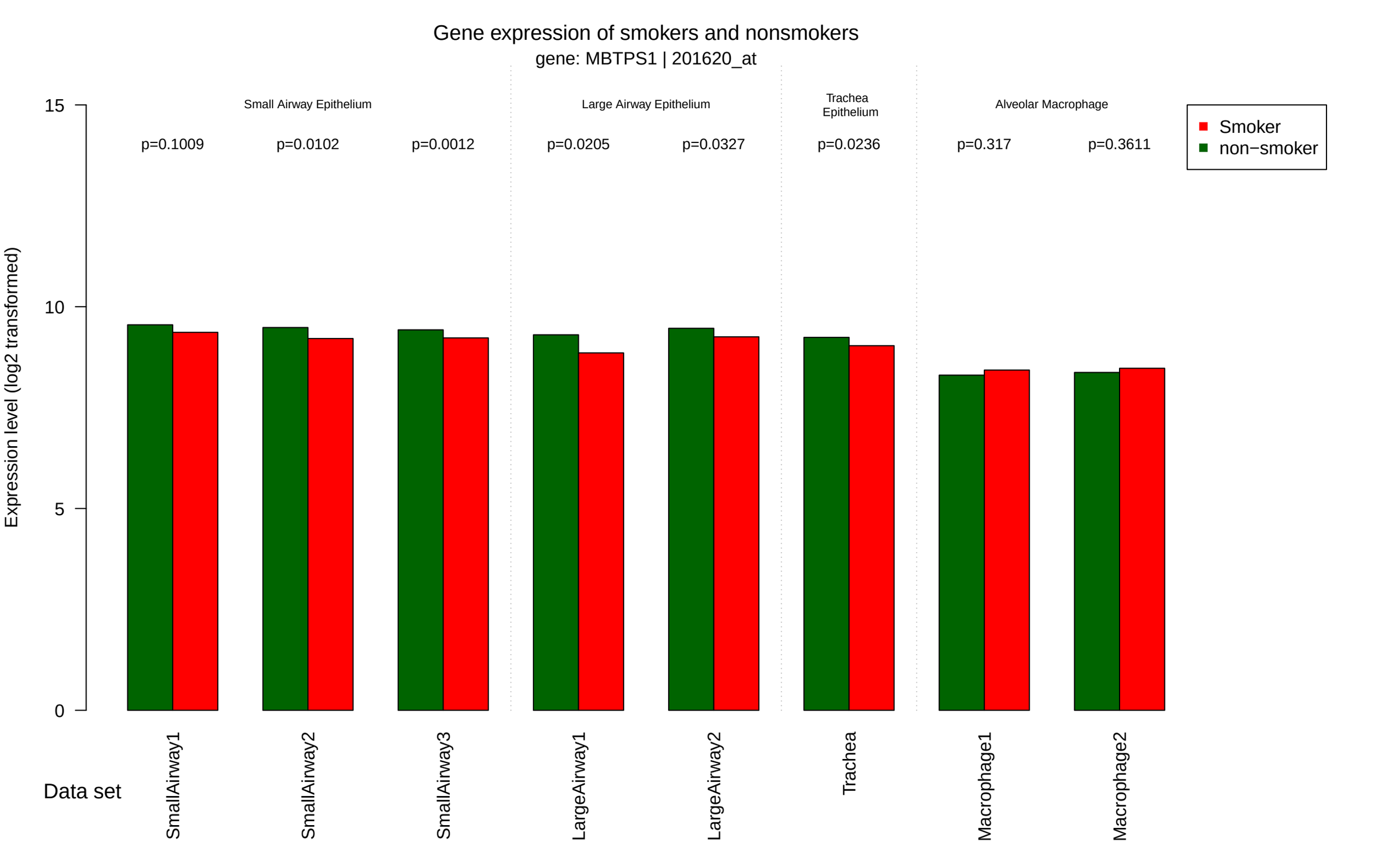
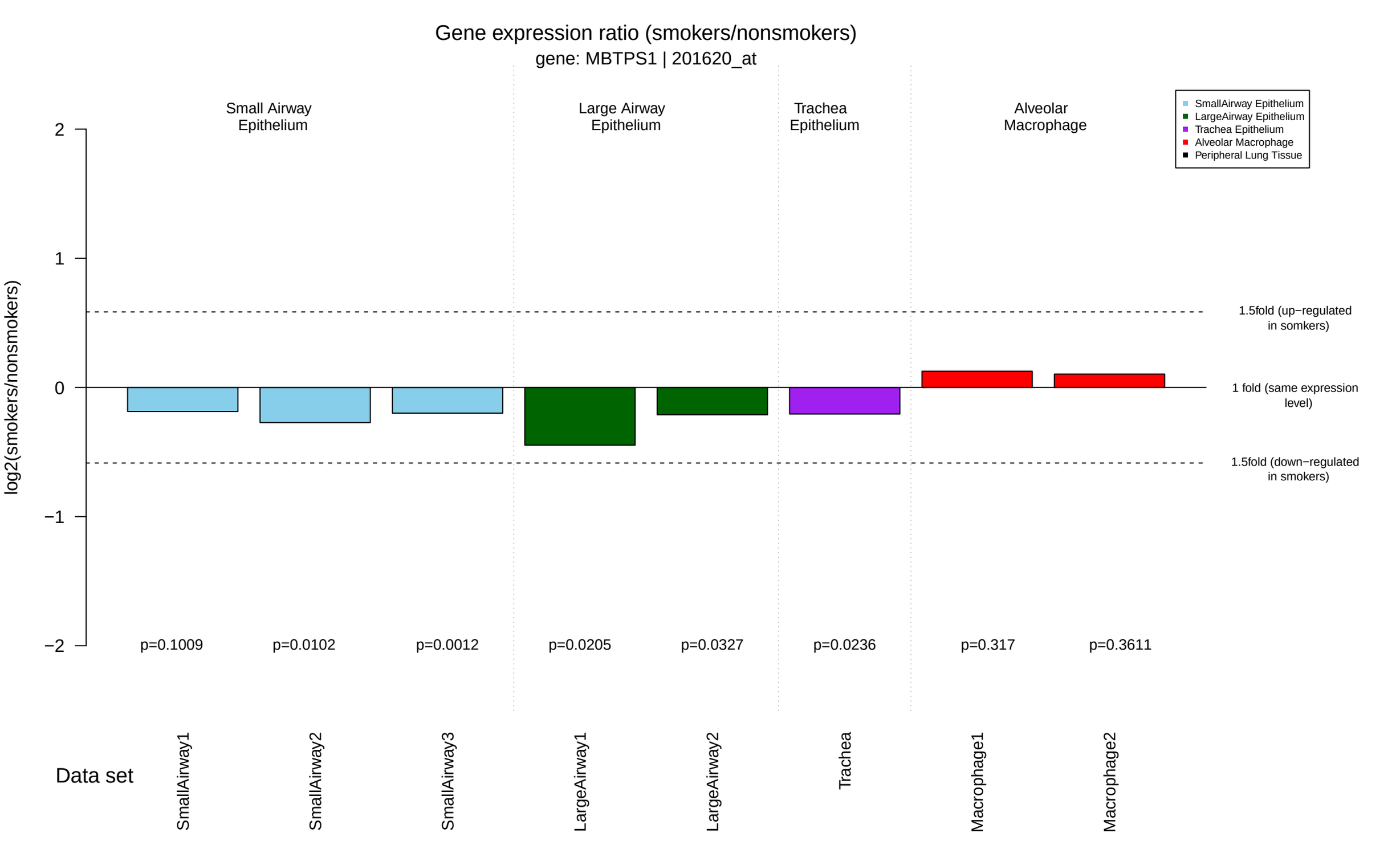
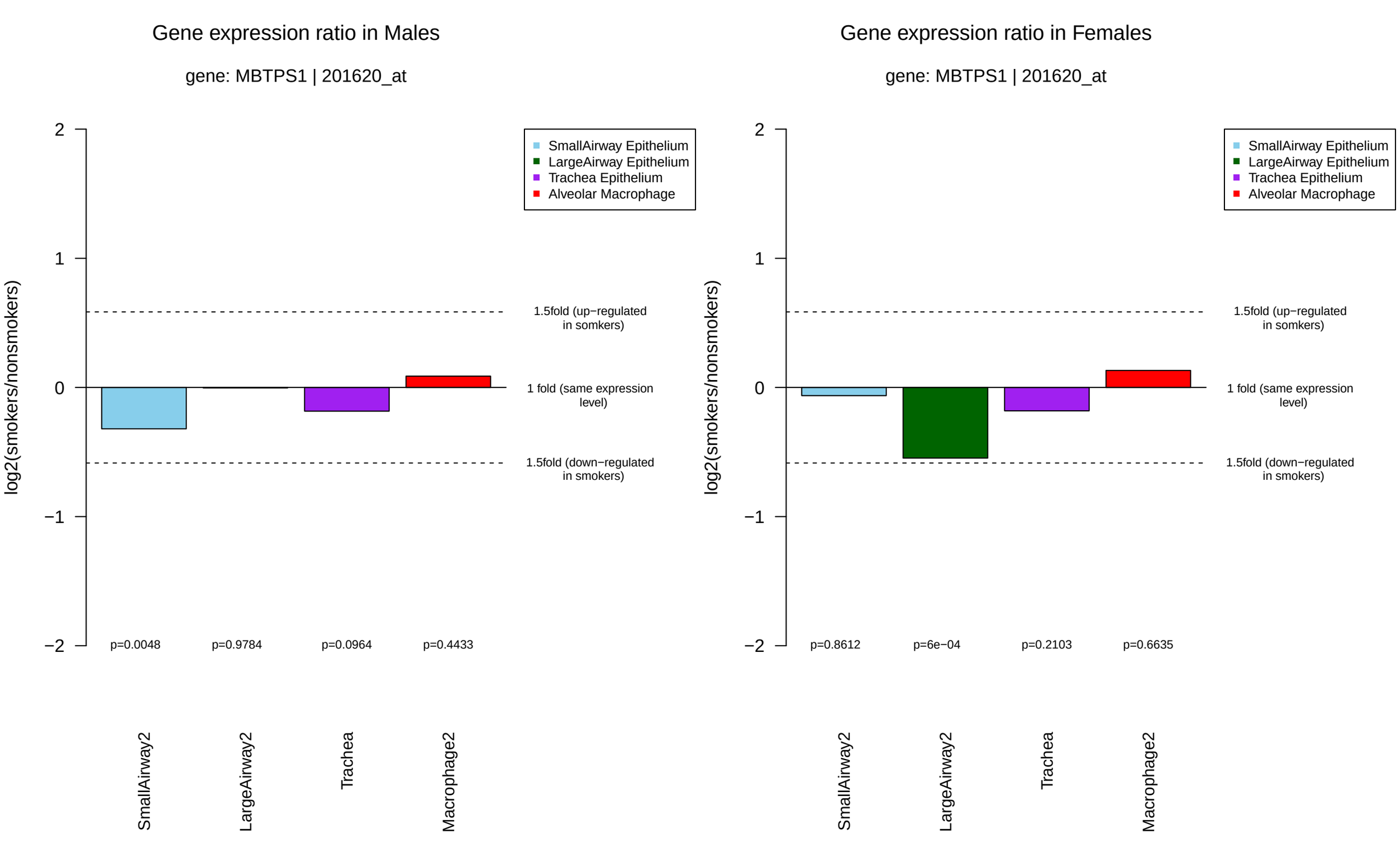
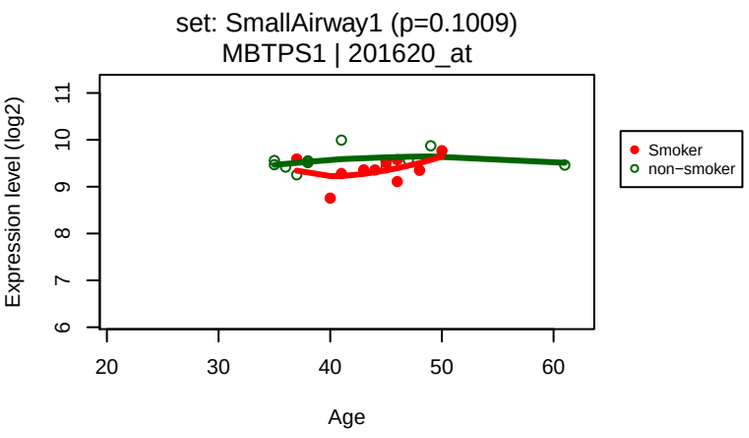
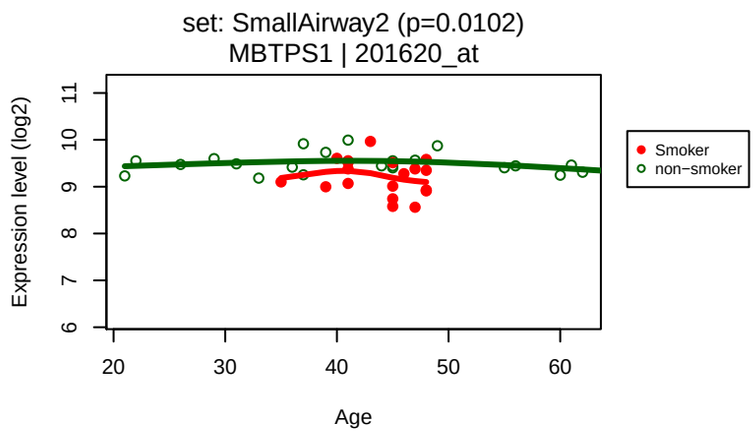
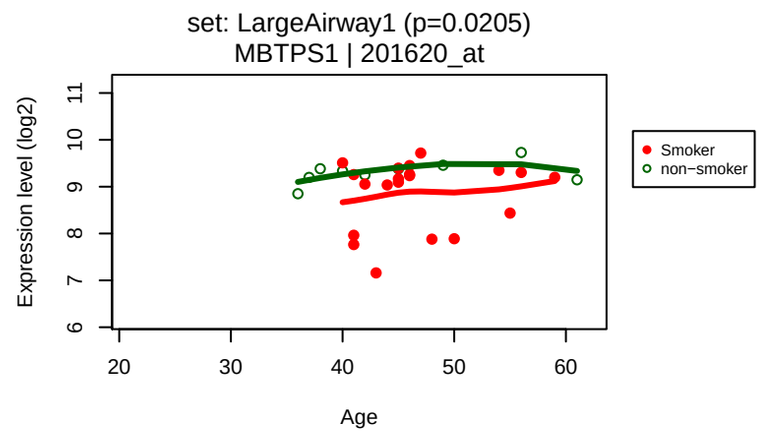

Gene expression of smokers and nonsmokers
gene: MBTPS1 | 201620_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1009
p=0.0102
p=0.0012
p=0.0205
p=0.0327
p=0.0236
p=0.317
p=0.3611
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

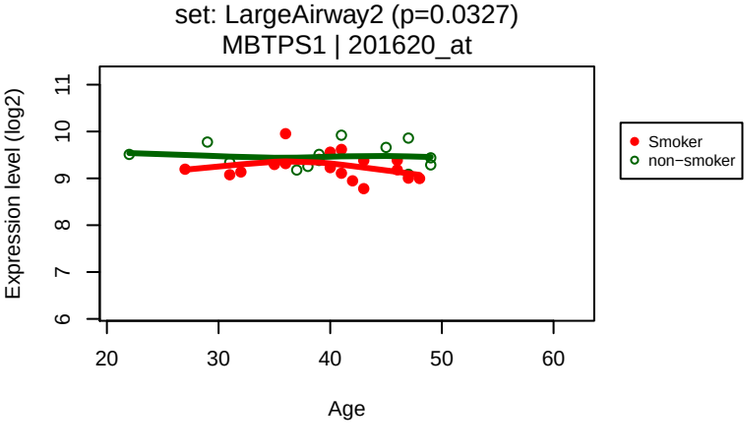

Gene expression ratio (smokers/nonsmokers)
gene: MBTPS1 | 201620_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1009
p=0.0102
p=0.0012
p=0.0205
p=0.0327
p=0.0236
p=0.317
p=0.3611
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

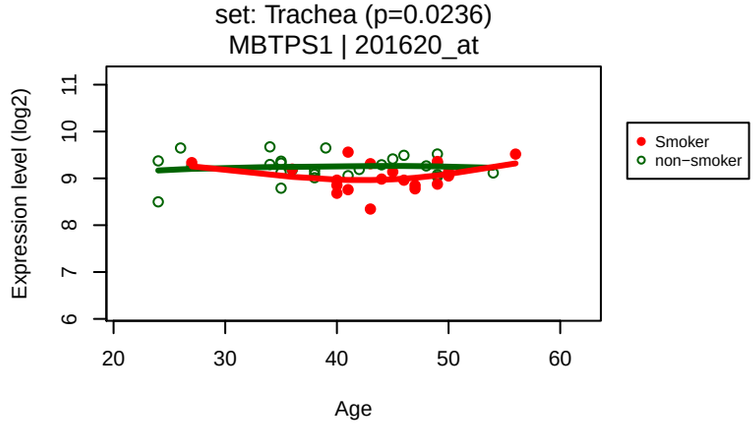

Gene expression ratio in Males
gene: MBTPS1 | 201620_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0048
p=0.9784
p=0.0964
p=0.4433
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MBTPS1 | 201620_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8612
p=6e−04
p=0.2103
p=0.6635
SmallAirway2
LargeAirway2
Trachea
Macrophage2

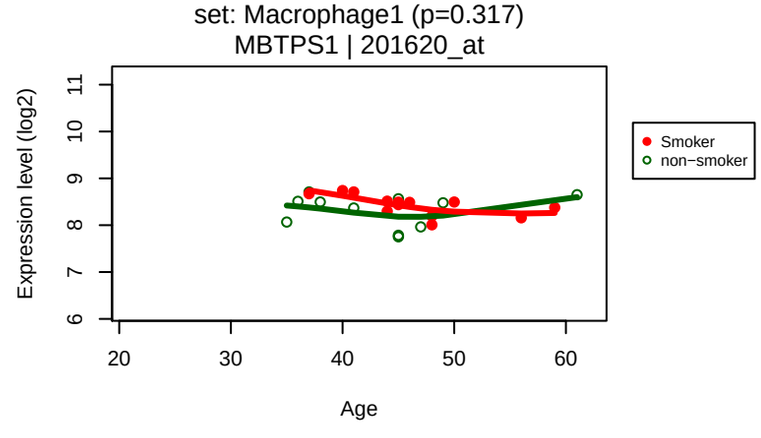

set: SmallAirway1 (p=0.1009)
MBTPS1 | 201620_at
Expression level (log2)
Age
Smoker
non−smoker

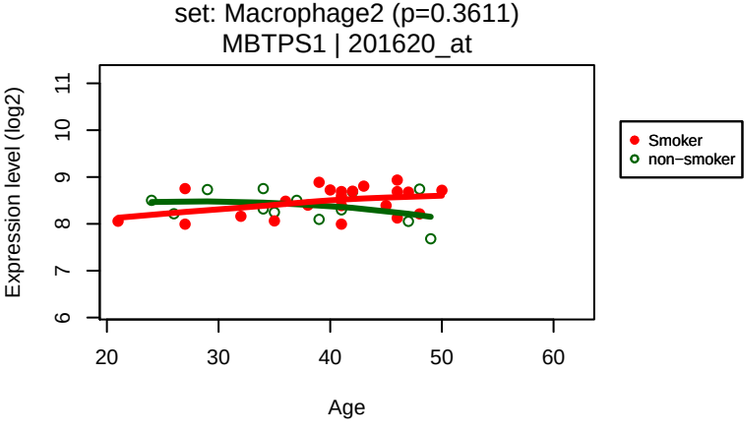

set: SmallAirway2 (p=0.0102)
MBTPS1 | 201620_at
Expression level (log2)
Age
Smoker
non−smoker

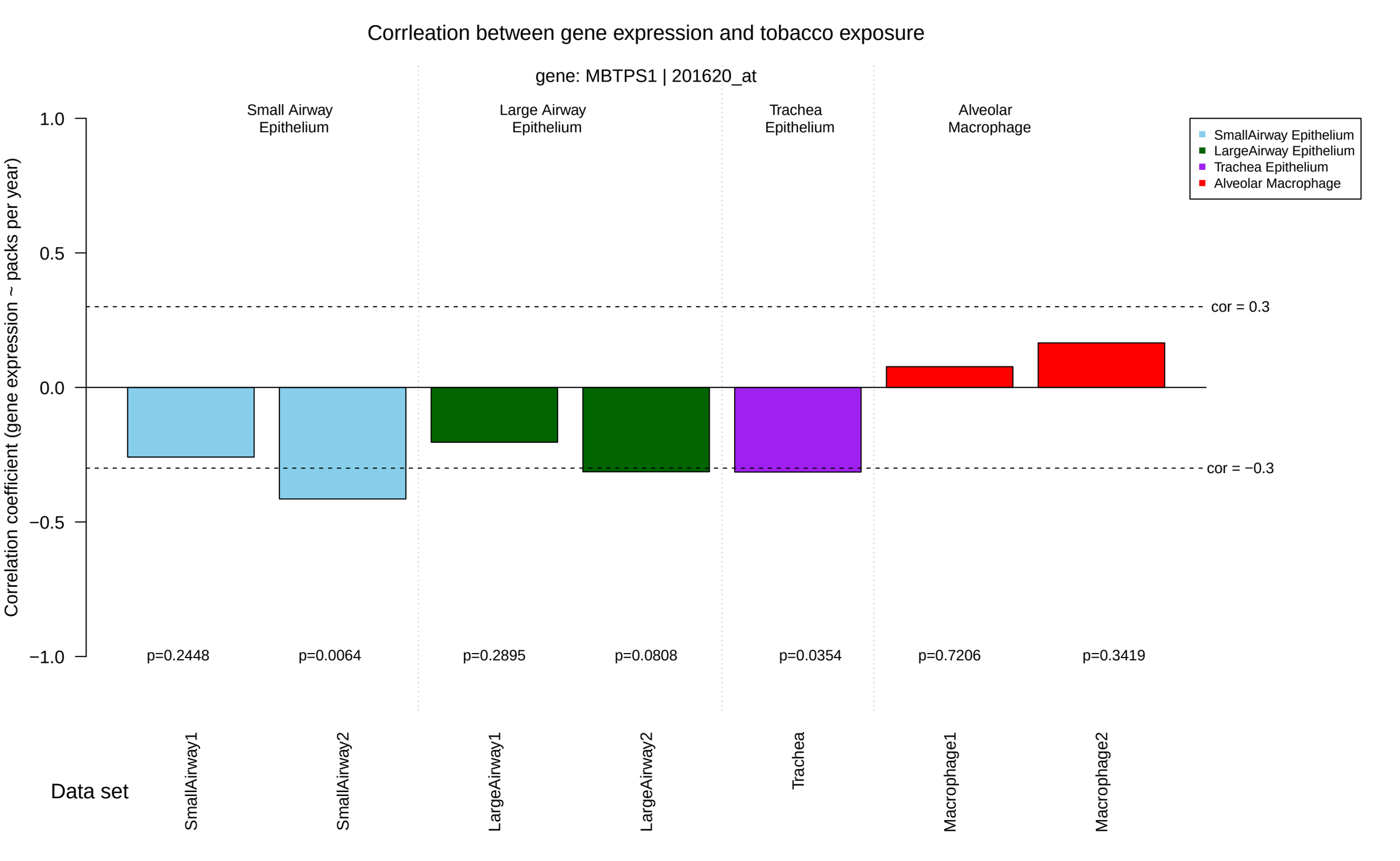

set: LargeAirway1 (p=0.0205)
MBTPS1 | 201620_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0327)
MBTPS1 | 201620_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0236)
MBTPS1 | 201620_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.317)
MBTPS1 | 201620_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3611)
MBTPS1 | 201620_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MBTPS1 | 201620_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2448
p=0.0064
p=0.2895
p=0.0808
p=0.0354
p=0.7206
p=0.3419
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2